

Dr. ASHWIN SUDHAKARAN

Physics – M.Sc., Ph.D., Post-Doc.,
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ABOUT ME

I am a dedicated physicist with a strong focus on material science & Electrochemistry, passionate about both research and education. With a solid academic background and notable achievements such as a Gold medal, I create an inclusive learning environment to equip students with skills for academic success and beyond. I contribute to cutting-edge research and strive to inspire a lifelong passion for learning and innovation, both as a researcher and educator.

EDUCATION

/Year of Completion

- **Post Doctoral Fellow (Material Science & Chemistry)** ...2024 - 2025
  CNMS, Jain University, Bangalore, Karnataka, India **(One year)**

  Overview of Research
  Independent researcher in the field of Material Science and Chemistry, on the topic of synthesis of bi-metallic and Electrospun ternary transition metal-oxide nanofiber catalysts for electrocatalytic CO_2 reduction and hydrogen evolution reactions (HER).

  Areas of exposure
  Electrospinning, Electrocatalyst, CO2RR, Multifunctional materials, Nanocomposites, Electrochemistry, Ferrites, Oxide materials, Transition metals, etc.

- **Ph.D., Physics (Material Science)** …2024
  Karpagam University, Coimbatore, Tamil Nadu, India.

  Dissertation Title: Synthesis and Characterization of Multiphase Barium Titanium - Zinc Ferrite Nanocomposites for Permanent magnet and Photonic applications.

  Degree Awarded (Convocation): July 26, 2025 **| Defended (Viva Voce):** August 19, 2024 **| Thesis Submitted:** May 25, 2023

Specialization: Physics (Material Science).

Thesis Overview: The thesis focuses on synthesizing and characterizing two types of Barium Titanate ($BaTiO_3$)-Zinc Ferrite (ZnFe2O4) nanocomposites using different preparation methods, including a chemical route and a green route.

The main objectives are,

- ✓ Synthesize $ZnFe_2O_4$ nanoparticles using chemical and green routes.
- ✓ Prepare two nanocomposites by mixing $BaTiO_3$ with $ZnFe_2O_4$ from each route.
- ✓ Characterize and compare the structural, morphological, spectroscopic, elastic, optical, dielectric, and magnetic properties of both composites.
- ✓ Examine the electron/charge density distribution and bonding theoretically to explain the properties.
- ✓ The nanocomposites are characterized using XRD, SEM, EDAX, FTIR, UV-Vis, PL, and VSM. Theoretical analysis of electron density distribution and bonding is done using Fullprof refinement data.

The results provide insights into how the synthesis method affects the overall properties of the nanocomposites. Key findings indicate that the green route composite exhibits superior crystallinity, whereas the chemical route composite possesses a higher optical bandgap. The composites exhibit soft ferromagnetic behavior, making them suitable for advanced permanent magnetic motor applications.

Overall, the research presents a quantitative analysis of how electron density differences resulting from varied synthesis methods affect the multifunctional properties of BaTiO3-ZnFe2O4 nanocomposites, with potential applications in permanent magnets and photonics.

Areas of exposure: Multifunctional materials, Metallic composites, powders, Nanocomposites, Ferrites, Oxide materials, Magnetic materials, Thin films, Phases & microstructural characterization, physical properties testing, heat treatment, etc.

- ➢ **M.Sc., Physics : 95.4 % (**Awarded **Gold Medal** - First class with distinction**)** …2019
  Karpagam University, Coimbatore, Tamil Nadu, India.

Dissertation Title: Quantitatively probing magnetic nanostructures $ZnFe_2O_4$ with superior spatial resolution

Project Institution: Indira Gandhi Center for Atomic Research (IGCAR), Department of Atomic Energy (DAE), Kalpakkam, Tamil Nadu, India.

Specialization: Physics (Material Science), Nanomaterials..

Thesis Overview: In my MSc thesis, I focused on the quantitative analysis of magnetic nanostructures, specifically ZnFe2O4, with an emphasis on

achieving superior spatial resolution. The key findings and conclusions of my research are as follows:

- ✓ I successfully synthesized phase-pure spinel nanostructured $ZnFe_2O_4$ using a co-precipitation method. This synthesis method ensured the desired phase and crystalline purity of the material.

- ✓ Comprehensive characterization techniques, including Raman spectroscopy and X-ray diffraction (XRD), confirmed the phase and crystalline purity of the $ZnFe_2O_4$ nanostructures.

- ✓ Microstructural analysis through scanning electron microscopy (SEM) and atomic force microscopy (AFM) revealed the smallest grain size, approximately 20 nm, which is crucial for understanding the material's properties at the nanoscale.

- ✓ Magnetic Force Microscopy (MFM) imaging unveiled the presence of very tiny magnetic domains, approximately 30 nm in size, within the $ZnFe_2O_4$ nanostructures. These images also showed variations in brightness, indicating differences in shape, size, and orientation of the nanostructures.

- ✓ A noteworthy observation was the relationship between magnetic domain intensity and grain size. It was found that as grain size increased, magnetic domain intensity increased and then decreased beyond a certain threshold. For grain sizes below 20 nm, MFM imaging exhibited significant diffusion of the phase signal, providing direct evidence for the presence of superparamagnetism in the nanostructured $ZnFe_2O_4$.

The results of this research demonstrate the significant impact of particle size on the magnetic properties of magnetic materials. This finding has promising implications for the development of novel magnetic materials for nano magnetic devices. Importantly, this work opens up the possibility of directly and quantitatively probing the weak magnetic stray fields emanating from nanoscale magnetic systems with unparalleled spatial resolution.

Areas of exposure: Nuclear physics, Nanomaterial characterization and sensors, Surface Nano scale examination, Material science.

- **B.Sc., Physics : 85.5 % (**First class with distinction**)** …2017
  Karpagam University, Coimbatore, Tamil Nadu, India

  Dissertation Title: Theoretical Study on LIGO (Laser Interferometer Gravitational Wave Observatory) as an Observatory

- **Higher Secondary (XII standard) : 86.67 %** …2014
  Hindusthan Matric Higher Secondary School, Coimbatore, Tamil Nadu, India.

  Subjects: Computer Science, Physics, Chemistry, Mathematics, Tamil, and English.

  Board of Studies: State Board of Tamil Nadu

- **Secondary School (X standard) : 85.4 %** …2012
  Adwaith G.N.S Matric Higher Secondary School, Coimbatore, Tamil Nadu, India.

  Subjects: Physics, Chemistry, Mathematics, Biology, Tamil, and English

  Board of Studies: State Board of Tamil Nadu

ACADEMIC / RESEARCH EXPERIENCE & RESPONSIBILITIES

- **Assistant Professor,** Department of Physics …04 Aug 2025 to Present
  Thiagarajar College of Engineering (TCE), Madurai, TN, India.

  As an Assistant Professor, I am actively engaged in teaching, academic administration, and research. Appointed in 16 June 2025 and officially joined on 04 August 2025, I serve as the coordinator for the first-year B.E. program and physics lab, and contribute to institutional accreditation processes. I have authored the Official Proposal to establish CoEMRES, which will coordinate the department's activities. I also handle responsibilities related to billing and stock maintenance (physics department). Alongside my academic duties, I am consistently involved in research publications and actively conduct workshops, hands-on training programs, and faculty development programs (FDPs) for the current academic year to foster academic excellence and knowledge dissemination.

  Responsibilities:
  - ✓ Department-level Coordinator (DLC), Centre of Excellence in Materials Research and Energy Solutions (CoEMRES)
  - ✓ DLC – Department of Physics, Extra-curricular activities
  - ✓ DLC - Documentation and compliance for NAAC, NIRF, NBA
  - ✓ First-Year B.E. Students Coordinator; Lab In-Charge
  - ✓ Departmental support: Bill processing, stock verification, general maintenance.
  - ✓ Anti-Ragging Squad Member for 1st Year BE Students.

  Other Leadership Roles:
  - ✓ Reviewer, Scientific Reports (Nature Portfolio), Journal of Sol-Gel Science and Technology, Optical and Quantum Electronics, Journal of Electroceramics (Springer Nature), 2024–Present.

- **Post Doctoral Fellow (Material Science & Chemistry)** …02 Sep 2024 to 02 Aug 2025 **(One Year)**
  CNMS, Jain University, Bangalore, Karnataka, India.

  As a Postdoctoral Researcher at the Center for Nano Materials and Material Science (CNMS), Jain University, Bangalore, I have been involved in research focused on the development of advanced nanostructured materials for electrocatalytic applications, particularly targeting CO_2 reduction and the hydrogen evolution reaction (HER). My primary responsibility has been to fabricate electrospun nanofiber-based catalysts tailored for enhanced

electrocatalytic activity. This work is currently ongoing and involves systematic optimization of electrospinning parameters, material synthesis, and subsequent physicochemical and electrochemical characterizations. The manuscript is in the final stages of preparation and will be submitted shortly. In parallel, I have contributed 02 review articles as a first author. Through this position, I have expanded my expertise in nanomaterials engineering, electrospinning techniques, and electrocatalytic system design, while also engaging in collaborative scientific writing and interdisciplinary teamwork.

Responsibilities:

- ✓ Characterized materials using XRD, FTIR, FESEM, EDX, and XPS
- ✓ Performed electrochemical studies for CO_2 reduction using a three-electrode system
- ✓ Co-authored research publications and assisted in manuscript preparation
- ✓ Mentored graduate students and contributed to research proposals
- ✓ Developed expertise in electrospinning and sustainable catalytic systems

➢ **Assistant Professor,** Department of Physics …15 May 2024
SRM MCET, Sivaganga, TN, India to
31 Aug 2024

At SRM Madurai College of Engineering and Technology (SRMMCET), I was involved in teaching core physics subjects to engineering graduates and actively contributed to curriculum development, student mentorship, and research supervision. My academic contributions extended to organizing scientific events and enhancing the institution's research profile through faculty training, student projects, and scholarly publications.

Responsibilities:

- ✓ Organized academic events: workshops, seminars, and development programs
- ✓ Supported academic audits and accreditation documentation (NAAC/NBA)
- ✓ Mentored students for higher education and competitive exams
- ✓ Contributed to lab development and interdisciplinary collaboration

Role: Head – R&D (Science & Humanities), SRMMCET:

- ✓ Promoted research activity among faculty and students
- ✓ Guided proposal writing and targeted SCI-indexed publications
- ✓ Mentored undergraduates in research basics and scientific reading
- ✓ Initiated development of an institutional academic journal

INTERNSHIP & TRAINING /Duration

➢ **Mendeley Training Course** …14/6/2020
Siva Institute of Pharmacy, Bilaspur, India.
Job position/Title: Trainee

Description: Mastering reference management and research workflow. Topics included software introduction, citation organization, and collaboration strategies. Acquired practical skills in document annotation and seamless citation insertion. Enhanced productivity and research efficiency.

- **M.Sc Final Year project Trainee** …SEP 2018 to APR 2019
  Indira Gandhi Center for Atomic Research (IGCAR),
  Department of Atomic Energy (DAE), Kalpakkam, Tamil Nadu, India.
  Job position/Title: Project/ Research Intern

  Description: Conducted research on ZnFe2O4 as part of M.Sc. project in collaboration with Karpagam University, TN, India, and developed training through hands-on experimentation and data analysis via Scanning Probe Microscope (SPM) - (Atomic & Magnetic Force Microscopy (AFM & MFM)

- **Summer Training Internship for Physics and Chemistry (STIPAC)** …28/5/2018 to 06/7/2018
  Indira Gandhi Center for Atomic Research (IGCAR),
  Department of Atomic Energy (DAE), Kalpakkam, Tamil Nadu, India.
  Job position/Title: Research Intern

  Description: Conducted research entitled, ''Nanoscale study of nanocomposite of BaTiO3x ZnFe2O4'' and exposed areas including Nuclear Physics, Nuclear power plant, Material Science, Nanomaterials, quantum optics, MEMS & NEMS.

FIELD OF INTEREST

- ✓ Material Science
- ✓ Electrochemistry
- ✓ Nanocomposites
- ✓ Multifunctional Materials
- ✓ Ferrites
- ✓ Metal Oxides
- ✓ Electrospun Materials
- ✓ Charge Density
- ✓ Microstructures
- ✓ Atomic and Molecular
- ✓ Condensed Matter
- ✓ Thin Films
- ✓ Green Synthesis
- ✓ Solid State

JOURNAL PUBLICATIONS

S. No.	Title	Journal Name / (National/ International)	Impact Factor / Indexing	Status / Vol, Year: Pg No.	Document Link / DOI
1.	Multifunctional insights into the Dy2FeMnO6 double perovskite: interplay of structural distortion, spin-polarized electronic structure, and optical properties.	Materials Advances/ International	5.8/ ESCI (Web of Science, Clarivate), Scopus	Published/ July 2026	View Article

2.	Comprehensive structural insights into Disordered Tb_2CoFeO_6 Double Perovskites for optoelectronic functionality	Vacuum/ International	4.3/ SCIE (Web of Science, Clarivate), Scopus	Published/ Vol 248, May 2026:115207	View Article
3.	Structure-property connections in dual-phase multifunctional barium titanate-based nanocomposites: Insights from charge density distributions	Materials Chemistry and Physics/ International	5.2/ SCIE (Web of Science, Clarivate), Scopus	Published/ Vol 318, May 2024:129290	View Article
4.	Multifunctionality of $AlBaFe_{12}O_{19}$ /$CoZnFe_2O_4$ hybrid nanocomposite: Promising structural, elastic, morphological, compositional, optical, and magnetic properties	Journal of Physics and Chemistry of Solids/ International	5.9/ SCIE (Web of Science, Clarivate), Scopus	Published/ Vol 174, Mar 2023: 111134	View Article
5.	Review: harnessing engineered electrospun materials for efficient CO_2 conversion into value-added products	Journal of Materials Science/ International	4.5/ SCIE (Web of Science, Clarivate), Scopus	Published/ Vol 60, May 2025: 8205–8220	View Article
6.	Optical Band gap characterization of Novel Nanoscale CBM- CZF composite	Bulletin of Environment, Pharmacology and Life Sciences (BEPLS)/ International	4.0/ Web of Science & UGC	Published/ Special Issue [1], May 2022 : 643-650	View Article
7.	Potential biological and optoelectronic applications of AgO: ZnO nanocomposite synthesized by green approach	The European Physical Journal Plus/ International	2.9/ SCIE (Web of Science, Clarivate), Scopus	Published/ / Vol 139 Issue 12, Dec 2024: 1118	View Article
8.	Investigating structure, elasticity, morphology, composition, and optical behavior of Al-doped $BaFe_{12}O_{19}$/$CoZnFe_2O_4$ hybrid composite	Journal of Materials Research/ International	2.9/ SCIE (Web of Science, Clarivate), Scopus	Published/ Vol 38, Jan 2023:1239 – 1253	View Article

9.	Electron density effect of hybrid ferrites on structural, spectroscopic, optical, and magnetic properties in $(BaTiO_3)_{(1-x)} + (ZnFe_2O_4)_x$ nanoceramics	Journal of Materials Science: Materials in Electronics/ International	3.2/ SCIE (Web of Science, Clarivate), Scopus	Published/ Vol 34, Jan 2023:104	View Article
10.	A Topological Control Algorithm for Decision-Making in Complex Material Systems	Indian Journal of Natural Science/ National	2.74/ Web of Science & UGC	Published/ Vol 15, Issue 85, Aug 2024: 76332-76342	View Article
11.	Ferrites: A study on structure, properties, synthesis, characterization, and Applications	Journal of Xidian University/ International	5.4/ Scopus & UGC Care	Published/ Vol 17, Issue 07, Aug 2023: 1474 -1526	View Article
12.	Investigating Optical Properties of Novel Heterostructure Aluminium Doped Barium Hexaferrite (AlBaM) Cobalt Zinc Ferrite (CZFO) Nanocomposite	Indian Journal of Natural Science/ National	2.74/ Web of Science & UGC	Published/ Vol 13, Issue 71, Apr 2022: 39368-39375	View Article
13.	Comprehensive Analysis of Optical, Dielectric and Electrical properties in Al- and Cu- Doped Barium Hexaferrite/Cobalt-Zinc Ferrite Hybrid nanocomposite	Journal of Electronic Materials/ International	2.8/ SCIE (Web of Science, Clarivate), Scopus	Published/ Vol 53, Issue 06, Jun 2024: 4622–4644	View Article
14.	Deliberating structural, optical, and spectroscopic properties of novel Barium Titanate Zinc Ferrite nanocomposite synthesized using low Temperature techniques	Stochastic modelling and applications/ National	1.35/ UGC Care Group I & Publon	Published/ Vol 26, Issue 03, Jan 2022: 598-605	View Article
15.	Studying structural and optical properties of novel Aluminium Barium Hexaferrite (ALBAM) Cobalt Zinc Ferrite (CZFO) nanocomposite	Stochastic modelling and applications/ National	1.35/ UGC Care Group I & Publon	Published/ Vol 26, Issue 03, Jan 2022: 606-612	View Article
16.	Qualitative Analysis of	International Journal	1.2/ Publon	Published/ Vol	View

	Magnetic Moment in $BaxTiO_3$ - $Ni_xFe_{3-x}O_4$ Nanocomposite Synthesized by Novel low temperature.	of Recent Technology and Engineering (IJRTE)/ International	(Web of Science)	10, Issue 05, Jan 2022 :17-22	Article
17.	Study of Bandgap Energy of Novel $(Cu_{0.5}Ba_{0.5}Fe_{12}O_{19})_{1-x}$/ $(Co_{0.6}Zn_{0.4}\ Fe_2O_4)_x$ Nanocomposite	International Journal of Recent Technology and Engineering (IJRTE)/ International	1.2/ Publon (Web of Science)	Published/ Vol 10, Issue 04, Nov 2021 :171-176	View Article
18.	Novel heterostructure $BaTiO_3$-$ZnFe_2O_4$ Nanocomposite : a Quantitative study	Rasayan Journal of Chemistry/ National	0.85/ Scopus & Web of Science	Published/ Vol 15, Issue 04, Dec 2022 :2197-2202	View Article

PUBLICATIONS IN CONFERENCE

S. No.	Title	Journal Name / (National/ International)	Indexing	Status / Vol, Year: Pg No.	Document Link / DOI
1.	Solar Energy Based LED Powered Light and Inverter Circuit with Different Types of Sensors	IOP Conference Series: Material Science and Engineering/ International	Scopus	Published/ Vol 1126, Mar 2021	View Article
2.	Ferrites: A study on structure, properties, synthesis, characterization and Applications	ISBN book Chapter in Conference Proceedings / International	---	Published/ OP-206, Jul 2021:301	View Article
3.	A Review on Factors affecting Structural, Morphological and Dielectric Properties of Barium Titanate Using Different Preparation Techniques	ISBN book Chapter in Conference Proceedings / International	---	Published/ OP-209, Jul 2021:304	View Article

INSTRUMENTS HANDLED

- ✓ Scanning Probe Microscopy (SPM: AFM and MFM Mode)
- ✓ Thermal Vaccum Evaporation Coating Unit (Thin Film Coating)
- ✓ Fourier Transform Infrared Spectroscopy (FTIR)
- ✓ CH Electrochemical workstation (3-electrode setup)
- ✓ Sophisticated single-needle Electrospinning
- ✓ Ultraviolet-visible Spectroscopy (UV-Vis)
- ✓ Photoluminescence (PL)
- ✓ Differential Scanning Calorimetry (DSC)
- ✓ Thermo Gravimetric Analysis (TGA)
- ✓ Simultaneous Thermal Analysis (STA)
- ✓ Gas Chromatography (GC)

SCIENTIFIC SOFTWARE USED

- ✓ Structure Refinement: Full-Prof, MAUD, GSAS-II, JANA 2006, HighScore Plus, Match-4, Profex, Pcpdfwin
- ✓ Bond Diagrams: VESTA, ChemDraw Ultra, Ortep, WInGX
- ✓ Fourier Transform Infrared Spectroscopy (FTIR).
- ✓ 3D CAD modeling and simulation: AutoCAD, FreeCAD, Fusion 360.
- ✓ FTIR Analysis: OMNIC, IRPal
- ✓ Morphology/ Texture: ImageJ
- ✓ Graph Plotting: Origin Lab, MAT LAB, QtiPlot
- ✓ Differential Scanning Calorimetry (DSC)
- ✓ DFT Stimulations: Materials studio (Biovia - CASTEP), Wein2k, Quantum ESPRESSO, COMSOL, OOMMF
- ✓ Plagiarism: Turnitin, Drill bit, etc.,

PRESENTATIONS

S. No.	Title	Place	Period	Remarks
1.	New Trends in advance functional materials	Online- SRM Institute of Science and Technology, Ramapuram, TN, India	31 July, 2025	**Participated** in this One day **International s**ymposium
2.	Theoretical and experimental investigation of multiferroic BaTiO3- ZnFe2O4 nanocomposite: Tailoring properties via green synthesis rote	Christ University sponsored by SERB & DRDO, India	18-20 April, 2024	**Presented (oral) in National Conference** on Sustainable energy conversion & storage: Functional materials, systems & Applications (NCSECS-2024)
3.	Insights into copper precursor	CNMS Jain University	10	**Presented (oral) in**

	influence on CuS catalyst synthesis for enhanced electrochemical CO_2 reduction into formic acid	sponsored by Rasayan Sanstha, India	October, 2023	**International Symposium** on Science beyond Boundary: Invention, Discovery, Innovation, and Society (Rasayan 17)
4.	Novel heterostructure BTO-ZFO nanocomposite: A Quantitative study	Centre for Material Science, Karpagam Academy of Higher Education, TN, India	June, 2022	**Presented in International Conference** on Recent Innovations in Multidisciplinary Scientific Research (ICRIMSR-2022)
5.	Analyzing magnetic properties of novel CBH-NZF nanocomposite	Centre for Material Science, Karpagam Academy of Higher Education, TN, India	June, 2022	**Presented in International Conference** on Recent Innovations in Multidisciplinary Scientific Research (ICRIMSR-2022)
6.	Quantitative analysis of low temperature Novel Heterostructure $BaTiO_3$-$ZnFe_2O_4$ Nanocomposites	Centre for Material Science, Karpagam Academy of Higher Education, TN, India	March, 2022	**Presented** in 1st **International Conference** on minerals, materials and manufacturing methods (ICMMMM-2022)
7.	Evidence of magnetic Interaction between $CuBaFe_{12}O_{19}$ and $CoZnFe_2O_4$ in the nanocomposite	Department of Physics, Faculty of Arts, Science, Commerce and management of Karpagam Academy of Higher Education, TN, India	May, 2022	**Presented in National Conference** on Advanced Materials and Its Applications (NCAMA-2022)
8.	Charge density analysis, structural, electrical and magnetic studies of $BaTiO_3$-$NiFe_2O_4$ heterostructure composite	Department of Physics, Faculty of Arts, Science, Commerce and management of Karpagam Academy of Higher Education, TN, India	May, 2022	**Presented in National Conference** on Advanced Materials and Its Applications (NCAMA-2022)
9.	Deliberating structural, optical and spectroscopic properties of novel Barium Titanate Zinc Ferrite	Online – Indian Academicians and Researchers Association	February 2022	**Presented** in 2nd **International conference** on “Science, Technology and innovation
10.	Review on magnetic and dielectric properties of $BaTiO_3$	SSN Research Centre-SSN Institutions, India	July, 2021	**Presented in International symposium** on “Modeling of Crystal Growth processes and Devices (MCGPD)
11.	Investigating optical properties of novel heterostructure Aluminium doped Barium Hexaferrite	Online – Indian Academicians and Researchers Association	February 2022	**Presented** in 2nd **International conference** on “Science, Technology and innovation

	(AlBaM) Cobalt Zinc Ferrite (CZFO) nanocomposite			
12.	High speed optical links and photonic devices for next generation data transfer	IIT Kota, India	11-15 October, 2021	**Presented** and participated in **National level 5 days workshop** sponsored by ATAL.
13.	Ferrites: A study on structure, properties, synthesis, characterization and applications	Online- Indian Association for Crystal Growth (IA- CG), held at Karpagam University	July, 2021	**Presented** in **International symposium** on "Modeling of Crystal Growth processes and Devices (MCGPD)
14.	Structural, morphological, magnetic and microwave absorption properties in Nickel Ferrite	Online- International Organization for Crystal Growth (IOCG),held at Karpagam University, TN, India	July, 2021	**Presented** in **International symposium** on "Modeling of Crystal Growth processes and Devices (MCGPD)
15.	A review on factors affecting structural, morphological and dielectric properties of Barium Titanate using different preparation techniques	Ramco Institute of Technology, Virudhunagar, TN, India	May, 2021	**Presented** in **National level Web Conference** on "Challenges and Innovations in Engineering and Technology (NWCCIET)
16.	Study on structure, properties, synthesis, characterization and applications of Ferrite materials	Ramco Institute of Technology, Virudhunagar, TN, India	February 2021	**Presented** in **National level Web Conference** on "Challenges and Innovations in Engineering and Technology (NWCCIET)
17.	Synthesis, characterization, properties and applications of $ZnFe_2O_4$ nanoparticles: A review	Karpagam University, Coimbatore, TN, India	May, 2021	**Presented** in **National Conference** on "Emerging Trends in Engineering Sciences (NCETES)
18.	Recent advancements in Hexagonal and spinel Ferrites	Online- Indian Association for Crystal Growth (IACG)	April, 2021	**Presented** in **International Virtual Conference** on "Energy Conversion and Storage (ICECS-2021)
19.	Structure-Property changes in Ferrites	Online- Indian Association for Crystal Growth (IACG)	April, 2021	**Presented** in **International Virtual Conference** on "Energy Conversion and Storage (ICECS-2021)
20.	New materials for a New Age - Multiferroics	Karpagam Academy of Higher Education, Coimbatore, TN, India	5-6 March, 2020	**Presented** this topic in **National Level seminar**
21.	Impact of Nanotechnology in Engineering Applications	Karpagam University, Coimbatore, TN, India	15 October,	**Participated** in a **National** level seminar

			2019	
22.	Advance materials and applications	Karpagam Academy of Higher Education, Coimbatore, TN, India	31 Jan to 1 Feb, 2019	**Presented** this topic in **National Level seminar**
23.	Nanoscale study of nanocomposite $BaTiO_{3-X}$ $ZnFe_2O_4$	Indra Gandhi Center for Atomic Research, IGCAR, Kalpakkam, TN, India	28 May to 06 July, 2018	**Presented and Attended National** level Summer Training Internship program in Physics and Chemistry (STIPAC) and carried out project work.
24.	Solid Oxide Fluid Cells (SOFC)	Karpagam University, Coimbatore, TN, India	20 February 2018	**Presented** this topic in **National Level seminar**
25.	Advanced materials	Karpagam College of Engineering, Coimbatore, TN, India	19-20 February 2017	**Participated** in a **National** level seminar
26.	Theoretical study on brief history of gravitational waves LIGO as an observatory	Karpagam University, Coimbatore, TN, India	February 2017	**Presented** this topic in **National Level seminar**
27.	Applications related to advanced materials	Karpagam Academy of Higher Education, Coimbatore, TN, India	31 Jan to 1 Feb, 2016	**Presented** this topic in **National Level seminar**

INVITED LECTURES / ACADEMIC OUTREACH

Chief Guest & Invited Awareness Lecture: *"Importance of Outcome-Based Education (OBE) in India and Student-Centered Learning"*, Annual Day Function, Government High School, V. Ramasamipuram, Vittalpatti, Sedapatti, Madurai, Tamil Nadu, India — 04 MAR 2026. (Offline).

Co-Chairperson: *"Second National level Research Conclave 2026"*, Thiagarajar College of Engineering, Madurai, Tamil Nadu, India — 26 to 27 MAR 2026. (Offline).

COURSES HANDLED

S. No.	Department	Course Title	Semester	Institution	Period
1.	B.E Mechanical Engineering	Physics (*OBE)	ODD, I SEM	Thiagarajar College of Engineering, Madurai, TN, India.	22 Aug to 18 Dec 2026 (Ongoing)
2.	B.E Electrical & Electronic	Physics (*OBE)	EVEN, I SEM	Thiagarajar College of Engineering, Madurai,	19 Jan to 22 May 2026

	Engineering			TN, India.	
3.	B.Tech Information Technology	Physics (*OBE)	EVEN, I SEM	Thiagarajar College of Engineering, Madurai, TN, India.	19 Jan to 22 May 2026
4.	B.Tech Computer Science and Business System	Fundamentals of Physics (*OBE)	ODD, I SEM	Thiagarajar College of Engineering, Madurai, TN, India.	22 Aug to 18 Dec 2025
5.	B.E Civil Engineering	Physics for Civil Engineering	ODD, I SEM	SRM Madurai College of Engineering and Technology, Sivagangai, TN, India	15 May to 31 Aug 2024
6.	B.E Electronics & Communication Engineering	Engineering Physics	ODD, I SEM	SRM Madurai College of Engineering and Technology, Sivagangai, TN, India	15 May to 31 Aug 2024
7.	B.Tech Information Technology	Engineering Physics	ODD, I SEM	SRM Madurai College of Engineering and Technology, Sivagangai, TN, India.	15 May to 31 Aug 2024
8.	B.E Computer Science	Engineering Physics	ODD, I SEM	SRM Madurai College of Engineering and Technology, Sivagangai, TN, India.	15 May to 31 Aug 2024

OBE = Outcome Based Education Practice

CONFERENCE/WORKSHOP/FACULTY DEVELOPEMENT PROGRAME (FDP)

S. No.	Title	Instituition	Period	Conference (or) Workshop (or) FDP Type/ Sponsor
1.	Advanced Functional Materials and emerging Technologies for Sustainable Energy, Electronics, Healthcare, Environmental Remediation, and Industrial Applications.	Thiagarajar College of Engineering (TCE), Madurai, TN, India	18-24 Nov, 2026	**About to conduct** One - week **International** FDP on the mentioned date as **Lead Organizer & coordinator**
2.	Magnetic Materials for renewable energy and storage systems	Thiagarajar College of Engineering (TCE), Madurai, TN, India	22-27 Oct, 2026	**Conducted** One-week **International** FDP on the mentioned date as **Lead Organizer & coordinator**
3.	Novel Functional Materials and their role in emerging technologies	Thiagarajar College of Engineering (TCE), Madurai, TN, India	19-25 Aug, 2026	**Conducted** One-week **International** FDP on the mentioned date as **Lead Organizer & coordinator**

4.	Advanced Functional Materials for Energy, Environment and sustainability	Online- SONA College of Technology, Salem, TN, India	27-31 July, 2026	**Participated** in this **One-week National** FDP.
5.	Unlocking Solar Hydrogen: Advancements and Challenges in PEC Water Splitting	Department of Chemistry, in association with the Centre for International Affairs (CFIA), Thiagarajar College of Engineering (TCE), Madurai, TN, India	31 July, 2026	**Participated** in this One day **International Research Seminar**
6.	Emerging Technologies for Clean Energy and Water Security	Online- Ethiraj College for Women, Chennai, TN, India	27-31 July, 2026	**Participated** in this **One-week Interdisciplinary** Virtual FDP.
7.	Additive Manufacturing & Advanced Nanomaterials	Online- EduCare Taiwan	24 July, 2026	**Participated** in this One-day **International webinar**
8.	Materials Insights: From Structural Refinement to Quantum Modelling	Thiagarajar College of Engineering (TCE), Madurai, TN, India	25-29 May, 2026	**Conducted** 5-days workshop (Hands-on) as **Lead Organizer & coordinator**
9.	Second National Level Research Conclave 2026	Thiagarajar College of Engineering (TCE), Madurai, TN, India	26-27 March, 2026	Served as a **member of Organizing committee** and supported to conduct the conclave as **CO-Chairperson**.
10.	International Conference on Materials for Electronics, Energy, Quantum and Biomedical Technologies (MQUBIT-2026)	Thiagarajar College of Engineering (TCE), Madurai, TN, India	12-13 March, 2026	As one of the **coordinators - Conducted this International Conference** with Various national and international speakers from esteemed universities, Institutions and Industries.
11.	Fundamentals and Practice of Rietveld Refinement using X-ray Diffraction data	Thiagarajar College of Engineering (TCE), Madurai, TN, India	25-26 August, 2025	**Conducted** 2-days workshop (Hands-on) as **Lead Organizer & coordinator**
12.	Emerging trends in Biomaterials, Materials design, and simulation	Thiagarajar College of Engineering (TCE), Madurai, TN, India	19 February 2026	**Conducted** 5-day FDP on the mentioned date as **Lead Organizer &**

	techniques			**coordinator**
13.	Advances in Computational and Experimental Research in Physics	SRM Institute of Science and Technology, Ramapuram, TN, India	21 July to 01 August, 2025	**Attended** Two-week **International** FDP
14.	Urban Photovoltaic Technology	Thiagarajar College of Engineering (TCE), Madurai, Tn, India in association with WRI & SEPA, India	25-26 June, 2025	**Offline** workshop/ Sustainability and Energy Practitioner Association (SEPA) an **Indo-German International** Collaboration
15.	Smart application development using IoT devices (FDP)	University of Engineering and Management (UEM), Jaipur, Kolkata, India	08 July, 2025	**IEM-UEM Online Training and Learning Platform** / UEM, India
16.	Design, synthesis and characterization of novel materials	Indian Institute of Information Technology (IIIT), Tiruchirappalli	25-29 October, 2021	**AICTE Training and Learning (**ATAL**)** / ATAL (FDP), India
17.	High speed optical links and photonic devices for next generation data transfer	Indian Institute of Information Technology (IIIT), Kota	11-15 October 2021	**AICTE Training and Learning (**ATAL**)** / ATAL (FDP), India
18.	Emerging research advances & trends in intelligent knowledge engineering systems	Sri Sai Ram Institute of Technology (Autonomous), Chennai	30 Sep to 06 Oct 2021	Six Days **International Level FDP/** Indian Society for Technical Education (ISTE)
19.	Future of examinations: Bloons taxonomy and multiple assessment tools	GTN Arts College (Autonomous), Dindigul	21-23 July, 2020	3 Days **National Level FDP**- Faculty Enrichment Program/ GTN Arts College (Autonomous), Dindigul
20.	Smart materials sensor and energy devices (SMSED - 2020	Karpagam University, Coimbatore, TN, India	25-30 May, 2020	**Attended** Six days **International Workshop**
21.	Solar thermal and photovoltaic systems	Karpagam Academy of Higher Education, Coimbatore, TN, India	Nov, 2017	**Participated** in two days **National** Workshop
22.	Science academies lecture workshop on Quantum mechanics	PSG Krishnammal College for Women, Coimbatore, TN, India	4-6 October, 2017	**Participated** in 3 days **National level** workshop sponsored by IAS Bangalore, INSA New Delhi, NASI Allahabad

ONLINE COURSES & CERTIFICATION

S.No.	Course Title	Learning Platform	Instructor	Period	Mode / Details
1.	Chemistry and Physics of Surfaces and Interfaces (Credit: 03, NCrF level 4.5-8.0)	NPTEL	Prof. Thiruvancheril G Gopakumar, IIT Kanpur, India	20 July - 11 Sep 2026 (8 Weeks)	**Online/**
2.	Instrumentation Program on Transition Electron Microscopy (TEM) and Fourier Transform Infrared Spectroscopy (FTIR)	Google Meet Link	Mrs. Nandhini Muthukumaran, Founder/ CEO, Nanonest Laboratory Pvt.Ltd, Coimbatore, TN, India	20-21 Nov, 2025	**Online/ Certificate ID:** NFT055
3.	Finite Element Analysis Convergence and Mesh Independence	Coursera	Jousef Murad, Mechanical Engineer, Coursera Instructor	20 June, 2025	**Online/** Link
4.	Environmental Science	IEM-UEM Teaching Learning	Dr. Ruchira Mukherjee, University of Engineering and Management (UEM), Jaipur, India.	18 June, 2025	**Online/ Certificate ID:** rzijiu1lkn
5.	Material Science and Engineering	IEM-UEM Teaching Learning	Dr. G.S. Taki, Department of ECE, Institute of Engineering and Management (IEM), Kolkata, India.	17 June, 2025	**Online/ Certificate ID:** qd265rda8t

PROFESSIONAL DEVELOPEMENT PROGRAME (PDP)

1. **Event Name:** AR/VR Wave is Going to be a Bigger Wave than the AI/ML Wave
   Speaker: Dr. M. Manivannan, Professor at IIT Madras, TN, India
   Date and Mode: 26th July 2024, (Online)

2. **Event Name:** Life @ TCE – A Faculty Orientation Program
   Speaker: Principal, Registrar, Dean and Faculties of Thiagarajar Engineering College (TCE), Madurai, India
   Date and Time: 16-22 July 2025, (Offline)

3. **Event Name:** 5th Faculty Conclave Life @ TCE – A Faculty developement Program
   Speaker: Faculties from EEE Department of Thiagarajar Engineering College (TCE), Madurai, India
   Date and Time: 12th February 2026, (Offline)

WEBINARS

INTERNATIONAL :
- Attended : 31
- Presented: 03

NATIONAL :
- Attended : 37
- Presented: 10

B O O K P U B L I C A T I O N S

1. **Sudhakaran et al.,** *Basics of major synthesis and characterization of nanomaterials*, Internationally published via Amazon Kindle Direct Publishing (KDP) platform, 2021. ISBN: 979-8723281752.
2. **Sudhakaran et al.,** *Fundamentals of zeolites & composites: nanoscience and nanotechnology*, Internationally published via Amazon Kindle Direct Publishing (KDP) platform, 2021. ISBN: 979-8746228833.
3. **Sudhakaran et al.,** *Chemistry behind nanoscience & nanotechnology*, Internationally published via Amazon Kindle Direct Publishing (KDP) platform, 2021. ISBN: 979-8806630927.
4. **Sudhakaran et al.,** *Basic fundamentals of graphene & graphene sheets based nanocomposites*, Internationally published via Amazon Kindle Direct Publishing (KDP) platform, 2021. ASIN: B094T2NTV5.
5. **Sudhakaran et al.,** *A review on mesoporous materials & their nanocomposites,* Internationally published via Amazon Kindle Direct Publishing (KDP) platform, 2021. ASIN: B094SQSX1C
6. **Sudhakaran et al.,** *Briefing various modeling techniques of polymer/CNT nanocomposites: Nanoscience & Nanotechnology*, Internationally published via Amazon Kindle Direct Publishing (KDP) platform, 2021. ASIN: B094R1RPDZ.
7. **Sudhakaran et al.,** *Introduction to cellulose, hemicellulose & lignin based nanocomposites: nanoscience & nanotechnology*. Internationally published via Amazon Kindle Direct Publishing (KDP) platform, 2021. ISBN: 979-8500770981.
8. **Sudhakaran et al.,** *An overview of CNT & their nanocomposites,* Internationally published via Amazon Kindle Direct Publishing (KDP) platform, 2021. ASIN: B094S92NY5
9. **Sudhakaran et al.,** *Briefing Metal-Organic Frameworks (MOF) & its composites*, Internationally published via Amazon Kindle Direct Publishing (KDP) platform, 2021. ISBN: 979-8500770981.

A W A R D S

- Awarded the **Gold Medal for achieving 95.4%** with **First Class with Distinction** during M.Sc. Physics at Karpagam University, Coimbatore, Tamil Nadu, India.

SKILLS

- **Analytical skills:** Conducted analysis of experimental data to investigate the effects of ferroelectric and magnetic phase coupling in multi-phase ferroic composites, using advanced spectroscopic and microscopy techniques, and applying quantitative analytical methods for data interpretation and validation.

- **Critical thinking:** Developed a new theoretical model to explain the coupling of ferromagnetic properties in multi-phase ferroic composites, providing new insights into the design of materials with enhanced multifunctional properties and critically evaluated existing models to identify limitations and propose improvements.

- **Research skills:** Conducted experiments to synthesize and characterize multi-phase ferroic composites and nanofibers with tailored compositions and microstructures, in order to optimize their multifunctional properties for use in a range of applications, including energy conversion and smart materials

- **Technical skills:** Proficient in using a range of laboratory equipment, including Thermal Vapour Evaporation Unit, UV, PL, FTIR spectroscopy, and advanced microscopy and spectroscopic instruments like SPM-AFM configuration. Experienced in electrochemical workstations (CH Instruments, Gamry), and thermal analysis (TGA/STA). Skilled in MAUD, FULLPROF and Origin for data analysis and modeling.

- **Communication skills:** Published research articles in high-impact scientific journals and presented research findings at international conferences in the fields of physics, materials science, electrochemistry and engineering. Actively engaged in technical writing, poster presentations, and scientific discussions.

- **Collaboration and teamwork:** Worked collaboratively with researchers from diverse backgrounds, including physicists, materials scientists, and engineers, on interdisciplinary research projects related to multi-phase ferroic composites in IGCAR, Kalpakkam and Metal Oxides with Mustansiriyah University and University of Baghdad, Iraq. Demonstrated strong interpersonal skills and adaptability in cross-cultural research environments.

- **Time management and organization:** Managed multiple UG and PG research projects simultaneously, ensuring that experimental timelines were met and research results were presented in a timely manner. Balanced administrative, teaching, and research responsibilities effectively.

- **Attention to detail:** Conducted careful characterization of the microstructure and properties of multi-phase ferroic composites and metal oxides, ensuring high-quality research outcomes and accurate reporting of research results.

- **Adaptability and flexibility:** Adapted research protocols to new experimental conditions and troubleshooting as needed, ensuring successful outcomes in the development and characterization of multi-phase ferroic composites and metal oxides. Quickly learned new tools, techniques, and instrumentation to meet evolving project demands

- **Intellectual curiosity:** Developed research questions and pursued innovative research directions in the fields of physics, material science, and electrochemistry, leading to publications. Continuously engaged in self-directed learning and literature review to stay current with emerging trends.

MEMBERSHIP IN PROFESSIONAL SOCIETIES

Name of the Society	Details of membership	Remarks	Nature of activities done through the society
International Association of Engineers (IAENG)	Lifetime member	Member ID: 303443	Published two articles in IJRTE journal suggested by IAENG
National Institute for Technical Training & Skill Development (NITTSD)	Lifetime member	Member ID: NITTSD/PROFESSOR/58855	
Institute For Engineering Research and Publications (IFERP)	Member	Member ID: PM47361890	Attended webinars suggested by IFERP
All India Council for Technical Skill Development (AICTSD)	Lifetime member	Member ID: AICTSD/PROFESSOR/72288	Attended webinars suggested by AICTSD
National Digital Library of India (NDLI) Club	Executive student member	Executive alumina member on behalf of the students of Karpagam Arts & Science college	Suggested ideas to improve Library facilities and infrastructure

COMPUTER PROFICIENCY

- ✓ Packages : Adobe Photoshop, Coral Draw, FlipaClip, Microsoft Office Suite, WPS Office, Libre Office, Latex.
- ✓ Languages : Basics in C++, C program
- ✓ Platform Used : Windows XP, Windows 7, Windows 10, Windows 11, Ubuntu and Linux.

CO-CURRICULAR ACTIVITIES

- Participated in Entrepreneurship Awareness Camp (EAC) (2016) organized by DST-NIMAT and Department of Commerce, Karpagam University, Coimbatore, TN, India.

LANGUAGES KNOWN

✓ English ✓ Tamil ✓ Hindi

EXTRA CURRICULAR ACTIVITIES

- Part of Guinness World Record participant creating world record in the Criteria, Largest one day Art competition (2010).
- Participated in Guinness book of world records and Limca Book of world record in "World's Longest Drawing" (2009).

WORK IN PROGRESS

- Writing scientific articles involving multifunctional transitional metal-based composites related to optical, magnetic, electrocatalytic CO_2RR and HER.
- Writing research articles related to Topological Edge states in CNT in collaboration with Dr. Raghavan Asokan and Dr. Harsha P, Department of Mathematics, Madurai Kamaraj University, Madurai.
- Writing articles related to Core/shell BTO-ZFO in collaboration with Dr. G. Mangamma Sairam, IGCAR, Kalpakkam.
- Writing articles related to Si doped magnetic materials in collaboration with Dr. Saranya Amirtharajan, Vel Tech Multi Tech Dr. Rangarajan Dr. Sakunthala Engineering College, Vel-Tech university, Chennai.
- Writing articles related to Zn doped materials for advanced antibacterial property in collaboration with Dr. Raghad S. Mohammed, University of Baghdad, and Mustansiriyah University, Iraq.
- Writing articles related to Tansition doped material for battery applications with Dr. Anila, Christ University, Bangalore, Karnataka, India.
- Writing articles related to rare earth doped double perovskite materials in collaboration with Dr. S. Rameshkumar, Vel Tech Multi Tech Dr. Rangarajan Dr. Sakunthala Engineering College, Vel-Tech university, Chennai.

REFERENCES

1. **Dr. M. Mahendran,**
   Prof., & Head, Department of Physics,
   Thiagarajar College of Engineering,
   Madurai-625015, Tamil Nadu, India.
   
   manickam-mahendran@tce.edu

2. **Dr. A. Saranya,**
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Vel Tech Rangarajan Dr. Sangunthala R&D Institute of Science and Technology,
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3. **Dr. Arvind H Jadhav**
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Centre for Nano and Material Sciences (CNMS)
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4. **Dr. S. Arumugam,**
Vice Chancellor,
Tamil Nadu Open University,
Saidapet, Chennai-600015, Tamil Nadu, India.
sarumugam1963@gmail.com

5. **Dr. G. Mangamma,**
Asso. Prof.: Homi Baba National Institute (HBNI) &
Scientific Officer H,
Head, Nanomaterials Characterization & Sensors Section
Surface and Nanoscience Division
Indira Gandhi Centre for Atomic Research (IGCAR),
Kalpakkam-603102, Tamil Nadu, India.
gm@igcar.gov.in

6. **Dr. P. Pandaram,**
Scientific Officer H, TLD Lab,
Kudakulam Nuclear Power project,
Kudakulam-627106, Tamilnadu, India
dr.p.pandaram@gmail.com

HOBBIES

- ✓ Drawing and Painting
- ✓ Reading
- ✓ Writing
- ✓ Meditation

DECLARATION

I, **Dr. ASHWIN SUDHAKARAN**, do hereby confirm that the information given above is true to the best of my knowledge.

Date:
Place:

(Dr. ASHWIN SUDHAKARAN)